

CHAPTER FIVE

NITRIFICATION

by

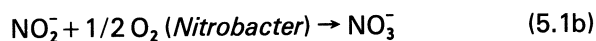
G.A. Ekama and G.v.R. Marais

1. INTRODUCTION

The term nitrification describes the biological process whereby free and saline ammonia is oxidized to nitrite and nitrate. Nitrification is mediated by *specific autotrophic organisms* with behavioural characteristics that differ significantly from the heterotrophic ones. The objectives in this chapter are to review briefly the kinetics of nitrification, to highlight the factors that influence this biological reaction and set out the procedure for designing a nitrifying-aerobic activated sludge process.

It has been well established that nitrification is due to two specific genera of autotrophic bacteria, the *Nitrosomonas* and *Nitrobacter*. Nitrification takes place in two sequential oxidation steps: (1) *Nitrosomonas* convert free and saline ammonia to nitrite, and (2) *Nitrobacter* convert nitrite to nitrate. The nitrifiers utilize ammonia for their synthesis nitrogen requirements, and as a source of energy to bring about synthesis. The ammonia requirement for synthesis, however, is a negligible fraction of the total ammonia processed by the organisms, at the most 2% (van Haandel and Marais, 1981). Consequently it is usual to neglect the metabolic nitrogen requirements* of the nitrifiers and to consider the nitrifiers simply to act as catalysts for the ammonia-nitrite-nitrate reactions – the reactions are taken as stoichiometric. This greatly simplifies the description of the kinetics of the process.

Consider the two basic redox reactions in nitrification,



Stoichiometrically the oxygen requirements for the first and second reactions are 3,43 and 1,14 mgO/mgN [also written as mgO/mg(NH₃-N)]. Hence the conversion of ammonia to nitrate, both expressed as N, requires 4,57 mgO/mgN utilized. Taking into account the ammonia utilized for metabolic purposes of the nitrifiers, the oxygen requirements per mg(NH₃-N) processed are slightly less, with reported values down to 4,3 mgO/mg (NH₃-N). However, such a sophisticated approach is more of academic than practical interest and we will accept the stoichiometric value, 4,57 mgO/mg(NH₃-N).

2. BIOLOGICAL KINETICS

2.1 Growth

In order to formulate the nitrification behaviour it is necessary to understand the basic biological growth kinetics of nitrification. Here one has to consider the two stage nitrification behaviour. The rate of conversion of ammonia to nitrate, by the *Nitrosomonas* is much slower than that of nitrite to nitrate, by the *Nitrobacter*. In fact, under most circumstances in municipal wastewater treatment plants, any nitrite that is formed is converted virtually immediately to nitrate. As a consequence very little nitrite is observed in the effluent from a plant operating on an influent that does not contain substances that inhibit the *Nitrobacter*. The limiting rate in the nitrification sequence, therefore, is that due to the *Nitrosomonas* – one needs to consider the kinetics of this organism only. Because the nitrite produced is virtually immediately converted to nitrate in terms of the growth rate of the *Nitrosomonas*, it can be assumed that the conversion is from ammonia to nitrate directly – on this basis the kinetics of nitrification reduce to the kinetics of the *Nitrosomonas*.

Experimental investigations by Downing, Painter and Knowles (1964) showed that the nitrification rate can be formulated in terms of Monod's relationship. Monod found that: (1) The mass of organisms generated is a fixed fraction of the mass of substrate (in this case ammonia) utilized. (2) The *specific rate of growth*, that is, the rate of growth per unit mass of organisms per unit time, is related to the concentration of substrate surrounding the organisms.

From (1):

$$M(\Delta X_n) = Y_n M(\Delta N_a) \quad (5.2)$$

where

$M(\Delta X_n)$ = mass of nitrifiers generated, (mgVSS)

$M(\Delta N_a)$ = mass of ammonia as N utilized, [mg(NH₃-N)]

Y_n = mass of nitrifiers generated per unit mass of ammonia utilized, [mgX_n/mg(NH₃-N)]

* In nitrification processes, where the influent energy source consists principally of ammonia, this simplification no longer can be applied and the determination of the nitrifying sludge mass is necessary to design the process volume (Neytzel-de Wilde, 1977). With municipal wastes under cyclic flow conditions the nitrifying mass also needs to be calculated.

Consequently one can write

$$\frac{dX_n}{dt} = Y_n \frac{dN_a}{dt} \quad (5.3)$$

From (2) Monod developed the following relationship:

$$\mu_{nT} = \frac{\mu_{nmT} N_a}{K_{nT} + N_a} \quad (5.4)$$

where

μ_n = specific growth rate observed at concentration N_a , (mg X_n /mg X_n /d)

μ_{nm} = maximum specific growth rate possible (mg X_n /mg X_n /d)

K_n = half saturation constant, i.e. the concentration at which $\mu_n = \frac{1}{2} \mu_{nm}$, [mg(NH₃-N)/l]

N_a = concentration of ammonia surrounding the organisms, [mg(NH₃-N)/l]

T subscript refers to temperature, °C

The *growth rate* is given by the product of the specific growth rate and the organism mass

$$\frac{dX_n}{dt} = \mu_{nT} \cdot X_n = \frac{\mu_{nmT} N_a}{K_{nT} + N_a} \cdot X_n \quad (5.5)$$

Instead of expressing the growth rate in terms of the mass of organisms, it is often useful also to express it in terms of the depletion rate of ammonia, as follows: From Eq. 5.3 substituting for dX_n/dt in Eq. 5.5

$$\frac{dN_a}{dt} = \frac{(\mu_{nmT}/Y_n) N_a}{K_{nT} + N_a} X_n \quad (5.6)$$

Equation 5.6 is sometimes written in an equivalent form as follows: Define the maximum specific substrate utilization rate, $K_{mT} = \mu_{nmT}/Y_n$ then the specific substrate utilization rate K_T becomes from Eq. 5.4

$$K_T = \frac{K_{mT} N_a}{K_{nT} + N_a} \quad (5.7)$$

and the rate of substrate utilization, from Eq. 5.6

$$\frac{dN_a}{dt} = \frac{K_{mT} N_a}{K_{nT} + N_a} \cdot X_n \quad (5.8)$$

This form is mentioned here because it is the one often used to express heterotrophic growth, and in fact is the form utilized in all the work published by the Marais group, because of its greater convenience for some purposes. The point to note is that Eqs. 5.6 and 5.8 are of equal merit.

The ammonia nitrogen that disappears in the nitrification reaction, reappears as nitrate nitrogen because the stoichiometric approach is adopted, i.e.

$$\frac{dN_n}{dt} = \frac{dN_a}{dt} \quad (5.9)$$

Consequently

$$\frac{dN_n}{dt} = \frac{(\mu_{nmT}/Y_n) N_a}{K_{nT} + N_a} \cdot X_n \quad (5.10)$$

where

N_n = nitrate concentration, (mg(NO₃-N)/l).

The oxygen consumption associated with nitrification is based on the stoichiometric assumption mentioned earlier, i.e. conversion of 1 mg(NH₃-N) to 1 mg(NO₃-N) requires 4.57 mgO. Consequently

$$\frac{dO_n}{dt} = 4.57 \frac{dN_a}{dt} \quad (5.11)$$

2.2 Endogenous respiration

Up to this stage attention has been focussed on the growth aspect. However, Herbert (1958) showed that generally an organism mass undergoes a continuous mass loss, called endogenous mass loss. Furthermore from the great amount of research undertaken by the Marais group it seems that this loss occurs independently of the growth. This allows straightforward formulation of endogenous mass loss as,

$$\frac{dX_n}{dt} = -b_{nT} X_n \quad (5.12)$$

where

b_{nT} = specific endogenous mass loss rate for *Nitrosomonas* (mg/mg/d)

2.3 Behavioural Characteristics

In Figure 5.1 the relationship between the specific growth rate, μ_n , the specific substrate utilization rate, K , and the substrate concentration, N_a is shown, as described by Monod's equation Eqs. 5.4 and 5.7 respectively. The rate constants selected are $\mu_{nm20} = 0.33$, $Y_n = 0.10$ (hence $K_m = 3.3$) and $K_{n20} = 1.0$. The interesting feature of this plot is that, because K_n is so small ≈ 1 mg(NH₃-N)/l, the specific rate is virtually at a maximum for concentrations of 2 mg(NH₃-N)/l and higher. However at concentrations less than two the rate rapidly declines to zero so that the ammonia is not readily reduced to zero.

3. PROCESS KINETICS

As indicated in Chapter 4 the basic process unit is the completely mixed activated sludge unit. The process under steady state provides the information necessary for design.

With regard to nitrification, for design, the principal steady state solution required is that of the effluent ammonia concentration from a completely mixed aerobic reactor. This solution forms the basis for extensive analysis of the process behaviour. A comprehensive analysis is not required here; for those interested reference should be made to Ekama, van Haandel and Marais (1979) and van Haandel and Marais (1981).

3.1 Effluent ammonia concentration

Consider nitrification in a completely mixed reactor, under constant flow and load conditions (see Figure 4.1). A mass balance on the accumulation of nitrifier mass $M(\Delta X_n)$ is given by

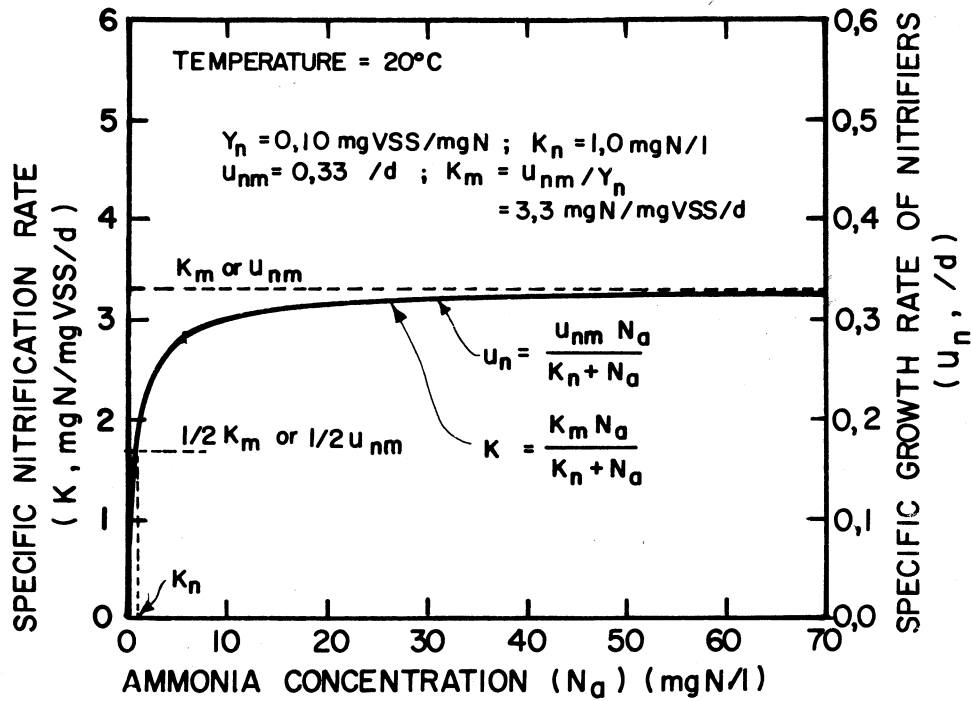


FIGURE 5.1: Monod's relationship for nitrification at 20°C.

$$M(\Delta X_n) = V_p \Delta X_n = [\mu_{nmT} \cdot N_a / (K_{nT} + N_a)] \cdot X_n \cdot V_p \Delta t - b_{nT} \cdot X_n \cdot V_p \cdot \Delta t - X_n \cdot q \cdot \Delta t$$

where

V_p = process volume (l)

q = volumetric withdrawal rate of waste sludge from the reactor (l/d)

Dividing by $V_p \Delta t$

$$\Delta X_n / \Delta t = [\mu_{nmT} N_a / (K_{nT} + N_a)] X_n - b_{nT} X_n - X_n q / V_p$$

Now

$$\frac{V_p X_n}{q X_n} = \frac{\text{Mass of nitrifier sludge in process}}{\text{Mass of nitrifier sludge wasted per day}} = \frac{V_p}{q} = R_s$$

where

R_s = sludge age (d) (see Eq. 4.2).

Under constant flow and load, when steady state is achieved, $\Delta X_n / \Delta t = 0$ and solving for N_a yields,

$$N_a = \frac{k_{nT} (b_{nT} + 1/R_s)}{\mu_{nmT} - (b_{nT} + 1/R_s)} \quad (5.13)$$

It is of interest to note that in Eq. 5.13 the ammonia concentration (N_a) in the reactor (and effluent) is independent of the specific yield constant (Y_n) and the influent ammonia concentration (N_{ai}). Using the same nitrification kinetic constants as those in Figure 5.1 and taking $b_{nT} = 0.0$, a plot of Eq. 5.13 with N_a versus sludge age R_s is given in Figure 5.2. Note that at long sludge ages N_a is very low and remains so until the sludge age is lowered to about 3.5 days below which, N_a increases rapidly until, in terms of this equation, N_a can exceed the influent concentration, N_{ai} —evidently the limit of application of the equation is when $N_a = N_{ai}$. Substitut-

ing N_{ai} for N_a in Eq. 5.13 and solving for R_s gives the minimum sludge age, R_{sm} , below which theoretically, no nitrification can be achieved. This minimum sludge age varies slightly with the magnitude of N_{ai} , see Figure 5.2—higher N_{ai} gives a *slightly* lower R_{sm} . Now the effective influent N_{ai} (see later) will rarely be less than about 20 mg/l. Noting that $K_{nT} \cong 1$ mgN/l, then

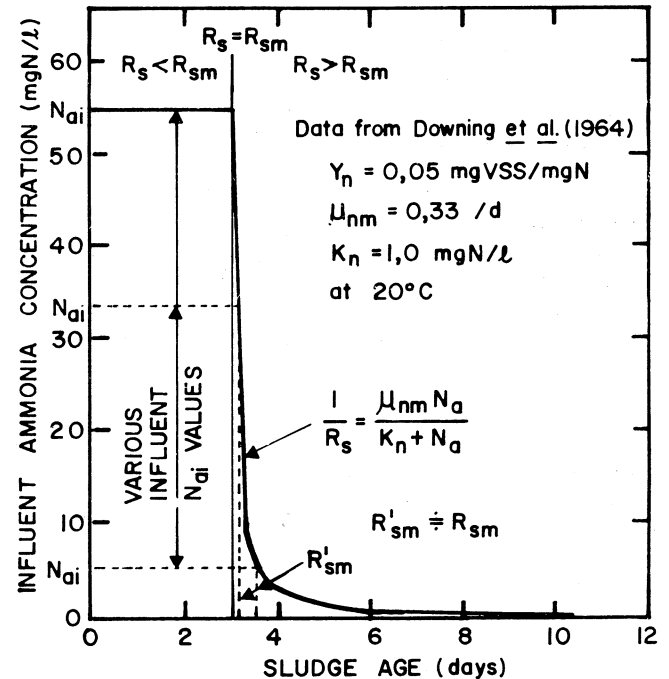


FIGURE 5.2: Relationship between sludge age for nitrification and influent ammonia concentration, N_{ai} .

K_{nT}/N_{ai} is negligibly small with respect to unity. Substituting zero for K_{nT}/N_{ai} in Eq 5.13 and solving for R_{sm} , yields

$$R_{sm} = 1/(\mu_{nmT} - b_{nT}) \quad (5.14)$$

For practical application, Eq. 5.14 adequately defines the minimum sludge age for all N_{ai} greater than about 5 mgN/l.

The virtually constant value for R_{sm} (for the accepted values of μ_{nmT} and b_{nT}) and the rapid attainment of high nitrification efficiency at sludge ages slightly greater than R_{sm} causes that in a particular plant, as the sludge age is increased, once R_s exceeds R_{sm} , high efficiency of nitrification will be observed. Consequently, under steady state conditions one would expect a plant either not to nitrify, or, to nitrify with high efficiency depending on whether R_s is shorter or longer than R_{sm} respectively. This behaviour is well attested by observation on laboratory scale plants.

From the discussion above clearly the magnitudes of μ_{nmT} and b_{nT} affect R_{sm} ; it is of importance, therefore, for practical application, to enquire into those factors that influence these constants and the nitrification phenomena in general.

4. FACTORS INFLUENCING NITRIFICATION

A number of factors affect the nitrification rate constants, efficiency of nitrification and the maximum sludge age. These are (1) wastewater source, (2) temperature, (3) pH, (4) unaerated zones, (5) dissolved oxygen concentration, and (6) cyclic flow and load.

4.1 Influent source

The maximum specific growth rate constant μ_{nmT} has been observed to be specific to the source of the waste flow and, even then, to vary between different batches from the same source. This specificity is so marked that μ_{nmT} should be classified as a wastewater characteristic. The effect appears to be of an inhibitory nature due to some substance(s) in the wastewater. It is not toxicity because high efficiency of nitrification can be achieved even with a low μ_{nm} value if the sludge age is increased sufficiently. These inhibitory substances are more likely to be present in effluents having some industrial waste components and in general, the higher this fraction, the lower μ_{nmT} tends to be, but the specific waste fractions that cause the reduction of μ_{nmT} have not been clearly delineated. Taking the μ_{nmT} value at 20°C as the reference value, μ_{nm20} values have been reported ranging from 0,33 to 0,65 per day. These two limits will have a significant effect on the minimum sludge age: Two plants, having these respective μ_{nm20} values, will have R_{sm} values differing by hundred per cent. Clearly due to the link between the waste flow and μ_{nmT} , the latter's value should always be estimated experimentally for optimal design. In the absence of such a measurement, a low value for μ_{nmT} necessarily will need to be selected which, for a particular waste flow, should the actual μ_{nm} be higher, will result in a non-optimal design. An experimental procedure to determine μ_{nm20} is given in Appendix 3.

With regard to b_{n20} , this value empirically is taken as constant for all waste flows, at $b_{n20} = 0,04/d$. Its effect is small so that there is no need to enquire closely into all the factors affecting it.

With regard to K_{nT} little information on effects of inhibitory agents is available; very likely K_{nT} will increase with inhibition.

4.2 Temperature

The μ_{nm} and K_n constants are sensitive to temperature with a high temperature coefficient (see Eq. 5.15); semi-empirically b_{nT} is taken also to have temperature dependency at the same rate as that for heterotrophs (see Eq. 5.16).

$$\mu_{nmT} = \mu_{nm20}(1,123)^{(T-20)} \quad (5.15a)$$

$$K_{nT} = K_{n20}(1,123)^{(T-20)} \quad (5.15b)$$

$$b_{nT} = b_{n20}(1,029)^{(T-20)} \quad (5.16)$$

The effect of temperature on μ_{nmT} is particularly dramatic, for every 6°C drop in temperature the μ_{nmT} value will halve which means that the minimum sludge age for nitrification will double. Design of plants for nitrification, therefore, should be based on the minimum expected process temperature. Note that the temperature sensitivity of K_{nT} does not affect the minimum sludge age, but it does affect the efficiency of nitrification – the higher the K_n the higher also the effluent ammonia at $R_s \gg R_{sm}$ (See Section 4.3).

4.3 pH and Alkalinity

The specific growth rate of the nitrifiers μ_n is extremely sensitive to the pH of the culture medium. It seems that the activities* of both the hydrogen and hydroxyl ions, (H^+) and (OH^-), act inhibitorily when their respective concentrations increase unduly. This happens when the pH increases above 8,5 [increasing (OH^-)] or decreases below 7 [increasing (H^+)]; optimal nitrification rates are expected for $7 < pH < 8,5$ with sharp declines outside this range.

From Eq. 5.4, μ_n is a function of both μ_{nm} and K_n . From an analysis of Eq. 5.13 it was shown earlier that the minimum sludge age is dominated by the magnitude of μ_{nmT} ; it is only very weakly influenced by K_{nT} . At $R_s \gg R_{sm}$ however, the effluent ammonia concentration (N_a), although low, is relatively speaking, significantly higher for larger K_{nT} values: For example if K_{nT} increases by a factor of two, the effluent ammonia concentration will increase correspondingly by the same factor (see Eq 5.13). Consequently the value of K_{nT} is significant in so far as it controls the effluent ammonia concentration at $R_s \gg R_{sm}$.

With regard to experimental estimates of μ_{nmT} and K_{nT} , a great amount of research has gone into tracing the effect pH has on μ_{nmT} . These investigations generally have not separated out the effects of μ_{nmT} and K_{nT} so that most data are in effect lumped parameter estimates of μ_{nmT} . Almost no information is available on the effect of pH on K_{nT} . The effect of pH on K_{nT} can be surmised from data obtained in a laboratory scale completely mix-

*Equivalent to concentration.

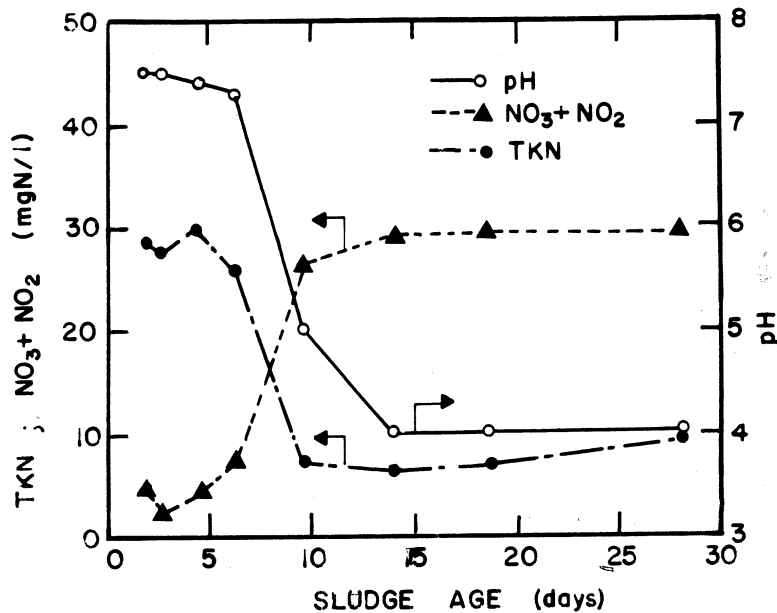


FIGURE 5.3: Relationship between TKN and nitrate concentrations and pH with sludge age in an activated plant treating weakly buffered influent sewage from the Strand-Somers West area, Cape.

ed reactor aerobic activated sludge investigation on a waste flow from Somerset West in the Cape Province. The plant was operated at 20°C over a range of sludge ages from 2 to 30 days. The response of the plant in terms of pH, filtered effluent TKN and nitrate plus nitrite is shown in Figure 5.3. Concomitantly with nitrification the pH declined, stabilizing at about 4 when the sludge age exceeded 15 days. The TKN stabilized at 10 mgN/l. Up to 10 days sludge age, the pH and nitrate concentration fluctuated in a see-saw fashion. The fact that stable nitrification was achieved only at 10 days sludge age when the pH was low, indicated that μ_{nmT} had decreased to a concomitantly low value. The fact that the effluent TKN ($\cong$ ammonia) stabilized at the high value of 10 mg/l indicated that K_{nT} had increased. On controlling the pH at 7.0 nitrification commenced at about 3 days sludge age and the effluent TKN ($\cong$ ammonia) stabilized at about 4 mg/l for $R_s > 3$ days. Evidently the pH and μ_{nmT} reacted interactively so that the minimum sludge age would fluctuate with the pH and achieve stability only at the long sludge age of 10 to 15 days. Evidently also the K_{nT} value is increased at low pH values. Due to the lack of information it is difficult to model quantitatively the effect of pH on μ_{nmT} and K_{nT} . The following analysis, based on various data sources and empirical assumptions on the behaviour, demonstrates the general expected trends even though the predicted values cannot be taken as quantitatively correct. Accepting that μ_{nm} remains constant for $7.2 < \text{pH} < 8.5$ and decreases as the pH decreases below 7.2 (Downing *et al.* 1964, Loveless and Painter, 1968) this behaviour can be modelled as follows:

For $7.2 < \text{pH} < 8.5$,

$$\mu_{nmpH} = \mu_{nm7.2} \quad (5.17a)$$

For $5 < \text{pH} < 7.2$,

$$\mu_{nmpH} = \mu_{nm7.2} \Phi_{ns}^{(\text{pH}-7.2)} \quad (5.17b)$$

where

$$\Phi_{ns} = \text{pH sensitivity coefficient} \cong 2.35.$$

In Figure 5.4 the predictions, using Eq. 5.17 are compared with experimental data on μ_{nm} collected by Malan and Gouws (1966). This comparison is illustrative only as it is not clear from the data of Malan and Gouws whether the effect of pH on K_n was taken into account when calculating μ_{nmpH} .

Changes in the value of K_n with pH are not at all documented in the literature. The best that can be done is to accept a dependency on pH and check the predictions against ammonia effluent concentrations reported at the different pH values. On this basis the following equations are suggested,

For $7.2 < \text{pH} < 8.5$

$$K_{npH} = K_{n7.2} \quad (5.18a)$$

For $5 < \text{pH} < 7.2$

$$K_{npH} = K_{n7.2} \Phi_{ns}^{(7.2-\text{pH})} \quad (5.18b)$$

where

$$\Phi_{ns} = \text{pH sensitivity coefficient} \cong 2.35.$$

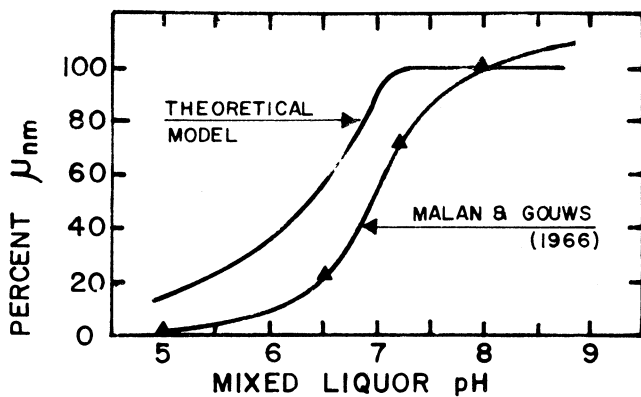


FIGURE 5.4: Relationship between maximum specific growth rate for *Nitrosomonas*, μ_{nm} , and mixed liquor pH. Experimental data after Malan and Gouws (1966).

The overall effect of pH or $5 < \text{pH} < 7.2$ and temperature on μ_n can be modelled by substituting Eqs. 5.15, 5.17 and 5.18 into Eq. 5.4 i.e.

$$\mu_{nT\text{pH}} = \mu_n \text{ at temperature } T^\circ\text{C and pH equal to pH.}$$

$$\mu_{nT\text{pH}} = \frac{\mu_{nm20}(2,35)^{(\text{pH}-7,2)}(1,123)^{(T-20)} \cdot N_a}{(K_{n20}(2,35)^{(7,2-\text{pH})}(1,123)^{(T-20)} + N_a)} \quad (5.19)$$

where

$\mu_{nT\text{pH}}$ = specific growth rate at temperature T and pH = pH

μ_{nm} = maximum specific growth rate at temperature 20°C and pH = 7,2

K_{n20} = saturation coefficient at temperature 20°C and pH = 7,2.

Although Eq. 5.19 is qualitative in prediction only, it serves an important function in illustrating the adverse effects that temperature and pH can have on μ_n . A plot of $\mu_{nT\text{pH}}$ is shown in Figure 5.5 for selected values of pH and temperature accepting $\mu_{nm20(7,2)} = 0,33/\text{d}$ and $K_{n20(7,2)} = 1,00 \text{ mgN/l}$. Clearly low pH and low temperature can have significant effects on design.

In design the temperature effects have to be accepted as restrictions imposed by the environment. With regard to the effects of low pH, these may, or may not, occur depending on the alkalinity of the influent. Low influent Alkalinity is likely to give rise to low pH in the purely aerobic activated sludge process. However, it is possible to limit, or completely obviate, pH reduction by operating the plant as an anoxic-aerobic process. The reasons for this are as follows:

From the overall stoichiometric equations for nitrification (Eqs. 5.1a and 5.1b), nitrification releases hydrogen ions which in turn decreases Alkalinity* of the mixed liquor. For every 1 mg ($\text{NH}_4^+\text{-N}$) that is nitrified 7,14 mg Alkalinity (as CaCO_3) is destroyed. Based on equilibrium chemistry of the carbonate system (Loewenthal and Marais 1977), equations linking the pH with Alkalinity for any partial pressure of carbon dioxide have been developed by van Haandel and Marais (1981). These relationships are shown plotted in Figure 5.6. When the Alkalinity falls below about 40 mg/l as CaCO_3 then, irrespective of the partial pressure of carbon dioxide, the pH becomes unstable and tends to decrease to low values. Generally, if nitrification causes the Alkalinity to drop below about 40 mg/l (as CaCO_3) problems associated with low pH very likely will arise in the plant, such as poor nitrification efficiency, tendency for developing bulking sludges, corrosive effluent, etc.

For any particular wastewater the possible effect of pH can be readily assessed, as follows: Take the exam-

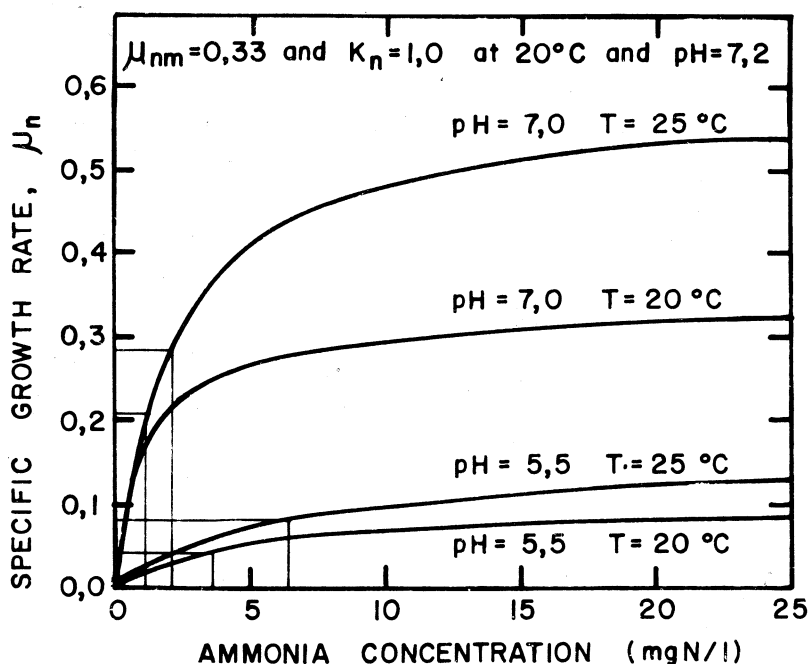


FIGURE 5.5: Diagrammatic representation of Eq. 5.19 - relationship describing the specific growth rate of *Nitrosomonas* for changes in temperature and mixed liquor pH.

*Alkalinity signifies the Total Alkalinity obtained by titrating the sample to an equivalent carbonic acid solution at pH approximately 4,3.

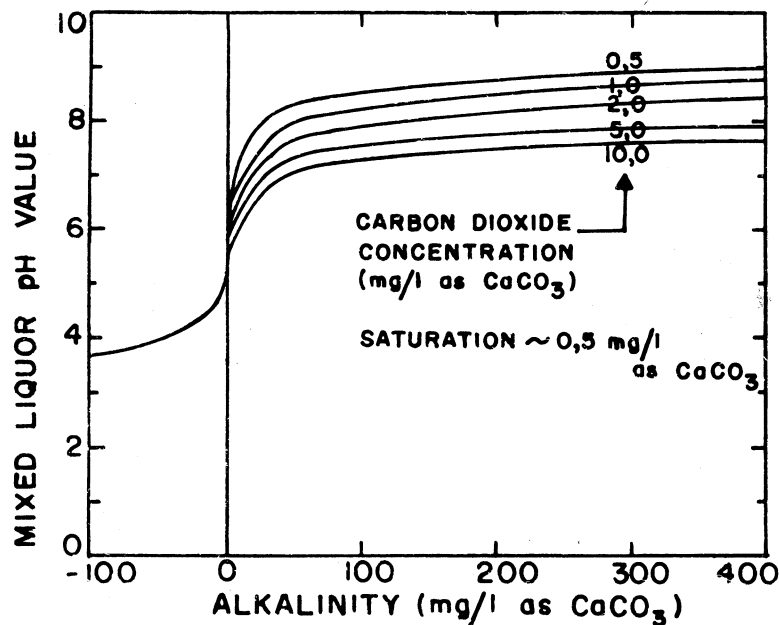


FIGURE 5.6: Mixed liquor pH versus alkalinity for different concentrations of carbon dioxide.

ple of an influent having an Alkalinity of 200 mg/l as CaCO_3 ; expected production of nitrate is 24 mgN/l. The expected Alkalinity in the effluent will be $(200 - 7,14 \cdot 24) = 29$ mg/l as CaCO_3 . From Figure 5.6, such an effluent is likely to have a pH < 7,0.

Wastewaters having low Alkalinity are often encountered where the municipal supply is drawn from areas underlain with sandstone.

The abovementioned problems are to be expected in aerobic nitrifying plants treating municipal effluents of towns located in the coastal areas of South Africa, from Cape Town to, and including, Natal. The effluent from Somerset West mentioned earlier (see Figure 5.3) is a typical example. The only practical approach to treating such effluents is to create an anoxic zone in the process to denitrify some or all of the nitrate generated. In contrast to nitrification, denitrification takes up hydrogen ions which is equivalent to generating Alkalinity. By considering nitrate as electron acceptor, it can be shown that for every mg ($\text{NO}_3\text{-N}$) denitrified, there is an increase of 3,57 mg Alkalinity as CaCO_3 . Hence incorporating denitrification in a nitrification process causes the net loss of Alkalinity to be reduced usually sufficiently to maintain the Alkalinity above 40 mg/l and consequently the pH above 7.*

In the example above, where the Alkalinity in the process is expected to decline to 29 mg/l as CaCO_3 , if 25% of the nitrate could be denitrified the gain in Alkalinity would be $(0,25 \cdot 29 \cdot 3,57) = 26$ mg/l as CaCO_3 and will result in an Alkalinity of $(29 + 26) = 55$ mg/l as CaCO_3 in the process. In this event the pH should remain above 7. For low Alkalinity wastewaters it is im-

perative, therefore, that denitrification be built into nitrifying plants.

Incorporation of unaerated zones in the process influences the sludge age of the process at which nitrification takes place so that cognizance must be taken of the effect of an anoxic or unaerated zone in establishing the sludge age of a nitrifying-denitrifying plant.

4.4 Unaerated zones

The effect of unaerated zones on nitrification can be readily formulated if the following assumptions are made:

- (1) Nitrifiers, being obligate aerobes, can grow only in the aerobic zones of a process.
- (2) Endogenous mass loss of the nitrifiers occurs under both aerobic and unaerated conditions.
- (3) The concentration of nitrifiers in the unaerated and aerated zones is essentially equal.

With these assumptions, Ekama, van Haandel and Marais (1979) showed that if a fraction f_{xt} of the total sludge mass is unaerated, i.e. $(1 - f_{xt})$ is aerated, the effluent ammonia is given by

$$N_a = \frac{K_{nT} (b_{nT} + 1/R_s)}{\mu_{nmT} (1 - f_{xt}) - (b_{nT} + 1/R_s)} \quad (5.20)$$

Equation 5.20 is identical in structure to Eq. 5.13, if one views the effect of the unaerated mass (f_{xt}) as

*Quantitative estimation of the denitrification that can be achieved in an anoxic-aerobic process is considered in Chapter 6.

reducing the value of μ_{nmT} to $\mu_{nmT}(1-f_{xt})$.

Following the same reasoning as that preceding Eq. 5.14 one can show that the minimum sludge age for nitrification R_{sm} in a process having an unaerated mass fraction, f_{xt} , is

$$R_{sm} = \frac{1}{\mu_{nmT}(1-f_{xt}) - b_{nT}} \quad (5.21)$$

Alternatively, if R_s is specified the minimum aerobic sludge mass fraction ($1-f_{xm}$) that must be present for nitrification is found by substituting R_s for R_{sm} and f_{xm} for f_{xt} in Eq. 5.21 and solving for $(1-f_{xm})$, i.e.

$$(1-f_{xm}) = (b_{nT} + 1/R_s)/\mu_{nmT} \quad (5.22)$$

or equivalently the maximum allowable unaerated sludge mass fraction is given by solving for f_{xm} in Eq. 5.22,

$$f_{xm} = 1 - (b_{nT} + 1/R_s)/\mu_{nmT} \quad (5.23)^*$$

For a fixed sludge age, R_s , the design value for the minimum aerobic sludge mass fraction ($1-f_{xm}$) always should be significantly higher than that given by Eq. 5.22, because nitrification becomes unstable when the aerated sludge mass fraction decreases to near the minimum value as given by Eq. 5.22. Consequently to ensure efficient nitrification (>90%) the *minimum aerobic sludge mass fraction* ($1-f_{xm}$) must be increased by a factor of safety, S_f to give the *minimum design aerobic sludge mass fraction*; from Eq. 5.22,

$$(1-f_{xm}) = S_f(b_{nT} + 1/R_s)/\mu_{nmT} \quad (5.24a)$$

and the corresponding *maximum design unaerated sludge mass fraction*, from Eq. 5.23 is

$$f_{xm} = 1 - S_f(b_{nT} + 1/R_s)/\mu_{nmT} \quad (5.24b)$$

With the aid of the temperature dependency equations for nitrification (Eqs. 5.15 and 5.16), a diagrammatic representation of Eq. 5.24 is given in Figure 5.7 for $S_f = 1,25$, $\mu_{nm20} = 0,65$ and $0,33/d$ at $14^\circ C$ and $20^\circ C$. The two μ_{nm20} values are the approximate extremes of the range of μ_{nm20} values reported in the literature. The diagram shows that f_{xm} is very sensitive to the maximum specific growth rate of the nitrifiers (μ_{nmT}): unless a sufficiently large *aerobic* sludge mass fraction is provided (i.e. $1-f_{xm}$, see right hand of Figure 5.7), nitrification will not take place and consequently nitrogen removal by denitrification is not possible.

If one fixes the minimum design aerobic sludge mass fraction, the sludge age can be determined by solving for $(1-f_{xm})$ in Eq. 5.24a. Knowing R_s the value of the effluent ammonia concentration (N_a) can be determined from Eq. 5.20. By doing a number of solutions of this kind, at $14^\circ C$, it will become apparent that if S_f is selected at 1,25 or greater the effluent quality will have stability and range from 2 to 4 mg (NH_3-N)/l depending on the other constants in the formulations. Accepting the sludge age as determined at $14^\circ C$, then at $20^\circ C$ the

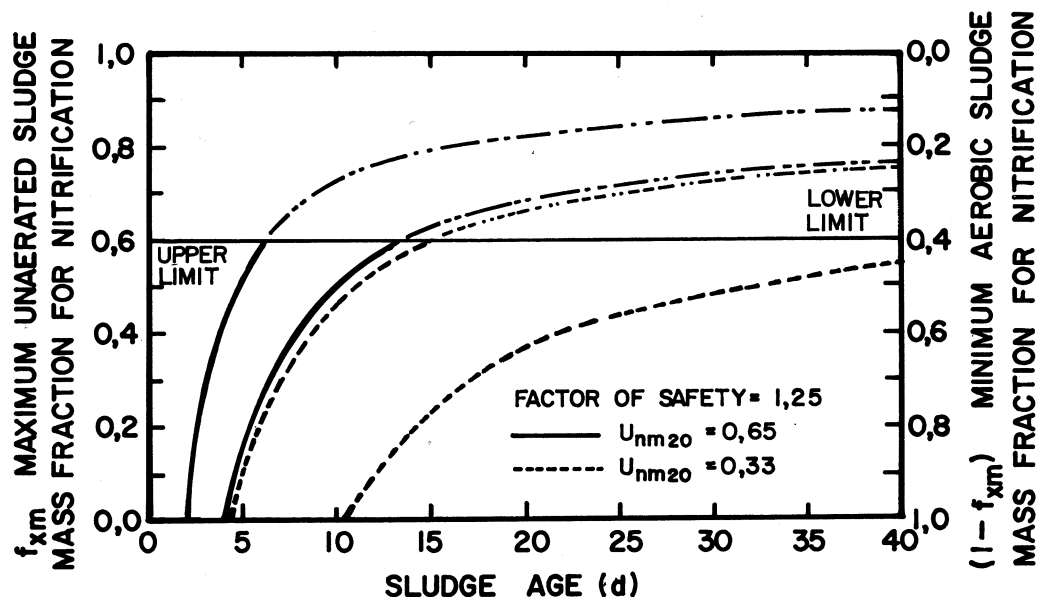


FIGURE 5.7: Maximum allowable unaerated sludge mass fraction and minimum aerated sludge mass fraction required to sustain nitrification versus sludge age for different maximum specific growth rates of the nitrifiers at $20^\circ C$ and $14^\circ C$ and $S_f = 1,25$.

*The usual approach for determining f_{xm} is based on the concept of "aerobic sludge age" i.e. equating the minimum sludge age for nitrification R_{sm} to the aerobic sludge age of the process R_{sa} . This leads to $f_{xm} = 1 - 1/\{R_s(\mu_{nmT} - b_{nT})\}$. This equation overestimates the maximum unaerated sludge mass f_{xm} at a particular sludge age because the approach ignores the endogenous mass loss of the nitrifiers in the unaerated zones. If the endogenous mass loss can be ignored in nitrification kinetics i.e. $b_n = 0,0$, then the two approaches are equivalent.

value of S_f will be found to have increased significantly and the effluent ammonia to have decreased to 2 mg/l or less. Consequently, for design the lower expected temperature should be selected to determine the sludge age and the aerobic mass fraction. If this is done, using $S_f = 1,25$, one can accept an effluent ammonia value of 2-4 mgN/l at the lowest temperature and 1-3 mgN/l at 20°C, that is, Eq. 5.20 is bypassed by the approximation above, provided R_s at least equals 1,25 R_{sm} . This approximation is sufficient for design and will be the approach followed in the design example.

4.4.1 Maximum allowable unaerated mass fraction

The equations above allowed the maximum design unaerated sludge mass fraction to be determined for any selected sludge age (Eq. 5.24b). If one accepts $S_f = 1,25$ then f_{xm} or $(1 - f_{xm})$ can be plotted versus R_s for any selected μ_{nmT} . This is illustrated in Figure 5.7 where the design (f_{xm}) and $(1 - f_{xm})$ values are plotted versus R_s for 20°C and 14°C for two selected μ_{nm20} reference values of 0,65 and 0,33 (b_{n20} was fixed at 0,04/d). Evidently from Figure 5.7, for $\mu_{nm20} = 0,65$ the unaerated mass fraction at 14°C can be approximately 0,8 at sludge age of 30 days. Such a high unaerated mass fraction is the indicated one theoretically at $R_s = 10$ days or longer at 20°C. One may now ask: Is this mass fraction acceptable for design or are other constraints present that may limit the fraction to lower values? To answer this, two aspects appear to be pertinent:

- (1) Experience with laboratory scale nitrification denitrification plants tends to support the opinion that at high unaerated mass fractions, the process is prone to bulking particularly at low temperatures. Processes at unaerated mass fractions of 0,5 or less show much less tendency in this regard.
- (2) An upper limit to the unaerated mass fraction is evident from experimental work on the Ludzack-Ettinger process, and theoretical predictions using the general activated sludge model. Experimentally at 20°C with $R_s = 20$ days, if $f_{xm} > 0,70$, the mass of sludge generated is found to increase sharply. This observation is indicated also by theoretical simulation. Theoretically, this happens for $f_{xm} > 0,60$ at $T = 14°C$ and $R_s = 20$ days. The reason is that for such high f_{xm} , the exposure of the sludge to aerobic conditions becomes insufficient to utilize the adsorbed and enmeshed particulate material. This leads to a decrease in active mass and oxygen demand and a build-up of enmeshed material, which causes a progressive decline in process activity because of the decline in active mass. When this happens, the process still functions in that the COD is removed from the wastewater, but the degradation of the COD is reduced; the process begins to assume the role of a contact reactor of a contact-stabilization process, i.e. a bio-flocculator with minimal degradation. This critical state occurs at lower f_{xm} as the temperature is decreased and the sludge age is reduced (Arkley and Marais, 1982).

From the discussion above it would appear that the unaerated mass fraction should not be allowed to increase above a certain upper limit of about 50%, but certainly not above 60%, as indicated in Figure 5.7.

In design of N and P removal plants the unaerated sludge mass fraction f_{xm} usually needs to be high i.e. $\geq 40\%$. If the μ_{nm20} value is low i.e. $< 0,40/d$ (which will be the usual case in design where insufficient information on the μ_{nm20} is available) the necessary high f_{xm} magnitudes will be obtained only at long sludge ages (see Figure 5.7). For example, if $\mu_{nm20} = 0,35/d$, then with $S_f = 1,3$ at $T_{min} = 14°C$, an $f_{xm} = 0,45$ (Eq. 5.24b) gives a sludge age of 25 days and for $f_{xm} = 0,55$ a sludge age of 37 days. Long sludge ages require large process volumes – increasing R_s from 25 to 37 days increases the process volume by 40% (see Figure 4.3) whereas f_{xm} increased only 22%. Also, for the same P content in the sludge mass, the P removal is reduced as the sludge age increases because the mass of sludge wasted daily decreases as the sludge age increases (see Figure 4.5). Consequently, for low μ_{nm20} values, the increase in N and P removal that can be obtained by increasing the unaerated sludge mass fraction above 0,50 to 0,60 might not be economical due to the large process volumes this will require, and might even be counter productive insofar as it affects P removal. A sludge age of 35 days probably is near the limit of economic practicality which, for low $\mu_{nm14} = 0,16$ values will limit the unaerated mass fraction to about 0,5. At higher μ_{nm14} values, the sludge ages allowing 50 per cent unaerated mass fractions decrease significantly again indicating the advantages of determining experimentally the value of μ_{nm20} to check whether a higher value is acceptable.

4.5 Dissolved oxygen concentration

High dissolved oxygen concentrations, up to 33 mg/l, do not appear to affect nitrification rates significantly. Low oxygen concentrations however do reduce the nitrification rate. Stenstrom and Poduska (1980) have suggested formulating this effect as follows:

$$\mu_{no} = \mu_{nmo} \frac{O}{K_o + O} \quad (5.25)$$

where

- O = oxygen concentration in bulk liquid (mgO/l)
- K_o = half saturation constant (mgO/l)
- μ_{nmo} = maximum specific growth rate (/d)
- μ_{no} = specific growth rate at O mg/l oxygen (/d).

The value attributed to K_o has ranged from 0,3 to 2 mgO/l, i.e. at DO values below K_o the growth rate will decline to less than half the rate where oxygen is present in adequate concentrations.

The wide range of K_o probably has arisen because the concentration of DO in the bulk liquid is not necessarily the same as inside the biological floc where the oxygen consumption takes place. Consequently the value will depend on the size of floc, mixing intensity and oxygen diffusion rate into the floc. Furthermore in a full-scale reactor the DO will vary over the reactor volume due to the discrete points of oxygen input (with mechanical aeration) and the impossibility of achieving

instantaneous and complete mixing. For these reasons it is not really possible to establish a generally applicable minimum oxygen value – each reactor will have a value specific to the conditions prevailing in it. In nitrifying reactors with bubble aeration a popular DO lower limit, to ensure unimpeded nitrification, is 2 mgO/l at the surface of the liquid.

Under cyclic flow and load conditions the difficulties of ensuring an oxygen supply matching the oxygen demand and a lower limit for the DO have been discussed previously (see Chapter 4, Section 11.3). Equalization of the load probably presents the best practical means to facilitate control of the oxygen level in the reactor. In the absence of equalization, mitigation of the adverse effects of low oxygen concentration during peak oxygen demand periods is by having sludge ages significantly longer than the minimum necessary for nitrification.

4.6 Cyclic flow and load

It is well attested experimentally that under cyclic flow and load conditions the nitrification efficiency of an activated sludge process decreases compared to that under steady conditions. This behaviour is also apparent in simulation studies using the general activated sludge model. From the simulation studies, during the high flow and/or load period, even though the nitrifiers are operating at their maximum rate, it is not possible to oxidize all the ammonia available, and an increased ammonia concentration is discharged in the effluent. This in turn reduces the mass of nitrifiers formed in the process – equivalently the effect of cyclicity in flow and load is to reduce the aerobic sludge age. The adverse effect of the cyclicity becomes more marked as the fractional amplitude of the flow and load cycles increase, and, is ameliorated as the aerobic sludge age increases. Simulation studies of the cyclic effect have indicated that an approximate relationship can be established between the effluent ammonia concentration under steady state and that under cyclic conditions (see van Haandel and Marais 1981). However if the factor of safety on nitrification, S_f , is 1,25 or greater the cyclic effect is relatively small and for design usually can be neglected.

At the risk of being tedious, it should again be pointed out that if the value of μ_{nm} is selected higher than the actual value, even with a sludge age of 1,25 to 1,35 times the theoretical minimum, the plant is likely to give rise to a fluctuating nitrate effluent with reduced mean efficiency in nitrification. Hence a conservative estimate of μ_{nm} is essential for a safe design.

Under cyclic flow and load conditions, it is necessary to know the value of both the specific yield coefficient of the *Nitrosomonas*, Y_n , the maximum specific growth rate constant, μ_{nmT} , and the half saturation constant, K_n ; in contrast, only μ_{nmT} and K_n are necessary in steady state calculations. The values of these constants have been determined by a variety of methods but the constants have a conjoined effect that may cause that in a particular method of evaluation different sets of values may be obtained. For practical pur-

TABLE 5.1 KINETIC CONSTANTS FOR *NITROSOMONAS* ACCEPTED IN THIS MONOGRAPH

Constant	Symbol	Value at 20°C	Temperature Coefficient	Eq No.
Specific yield coefficient	Y_n	0,10	1,000	—
Endogenous respiration rate	b_n	0,04	1,029	5.16
Half saturation coefficient	K_n^*	1,00	1,123	5.15b

*Note that K_n increases with increase in temperature.

poses it is not possible to obtain a complete set of values so that one approach is to accept the value of one constant and in any particular experimental evaluation determine the other constants from the experimental data using this value. The estimates for Y_n and K_n accepted in this monograph are listed in Table 5.1; experimental μ_{nm} determination, therefore, is relative to the value of Y_n and K_n listed.

5. DESIGN CONSIDERATIONS

5.1 Fate of the influent TKN

The various fractions of the influent TKN have been described in Chapter 2. Of these the following are of importance to our discussion: (1) free and saline ammonia concentration (N_{ai}); (2) biodegradable organic nitrogen (N_{oi}); and (3) the soluble unbiodegradable organic nitrogen (N_{ui}).

The biodegradable organic nitrogen is decomposed in the process by heterotrophic action to free and saline ammonia. Furthermore, on death of the organisms, by endogenous mass loss, the nitrogen released is in the organic bound form and the biodegradable fraction also is broken down to free and saline ammonia. These two sources of free and saline ammonia together with that in the influent are utilized by the organism mass first for the formation of new organisms and the balance for generation of nitrate, if the conditions are favourable.

The 'exact' formulations describing the interactions between the free and saline ammonia and the organic bound nitrogen are complex and can be developed effectively (at present) only in terms of the growth-death-regeneration model of Dold, Ekama and Marais (1981).^{*} However under steady state conditions the kinetics simplify and the formulations reduce to describing net effects. These can be developed readily in terms of the synthesis – endogenous respiration approach already used to develop the steady state equations for carbonaceous material degradation, in Chapter 4. For design there is even a further simplification possible: In nitrification systems if, for example, the constraints are put on the system that the minimum design sludge age must be at least 1,25 to 1,35 times the theoretical

^{*}A more elaborate discussion is available by van Haandel and Marais (1981).

minimum to initiate nitrification, and, that the pH remains above 7, some of the steady state kinetic equations predict effluent concentrations that fall within a narrow band of values, sufficiently narrow that one can accept the effluent parameter virtually as constant.* Below are set down some of these steady state equations of importance in developing design procedures. Simple approximations adequate for design are indicated and some general response graphs are derived.

In the design of those aspects of a process effecting nitrification, basically the following information is needed: (1) effluent TKN concentration; and (2) effluent nitrate concentration.

5.2 Effluent TKN

The effluent TKN will consist of free and saline ammonia (N_a), biodegradable soluble organic N (N_o), unbiodegradable soluble organic N (N_u) and TKN in the effluent volatile solids. Equations for the fractions are set out below:

(1) Free and saline ammonia (N_a):

N_a is given by Eq. 5.20, i.e.

$$N_a = \frac{K_{nT} (b_{nT} + 1/R_s)}{\mu_{nmT} (1 - f_{xt}) - (b_{nT} + 1/R_s)} \quad (5.20)$$

Note that N_a is independent of the influent ammonia concentration. Note also that the equation applies only if the sludge age $R_s > R_{sm}$, however this condition always should be satisfied in design.

(2) Biodegradable organic nitrogen concentration (N_o)

The biodegradable organic nitrogen is broken down to free and saline ammonia by heterotrophic organisms. Although a complex reaction, involving a feedback of this nitrogen form from death of the organisms, the prediction simulated by the general model indicates that the following equation, in terms of the influent biodegradable organic nitrogen concentration, is adequate

$$N_o = \frac{N_{oi}}{1 + K_r X_a R_{hn}} \quad (5.26)$$

where

N_{oi} = influent biodegradable organic nitrogen mg(TKN-N)/l. Its value is given by Eq. 2.14.

K_r = kinetic constant for degradation of N_o . Its value and temperature dependency is given in Table 5.3.

X_a = active mass concentration in aerobic reactor.

R_{hn} = nominal hydraulic retention time in process.

(3) Soluble unbiodegradable organic nitrogen (N_u)

$$N_u = N_{ui} \quad (5.27)$$

where

N_{ui} = soluble unbiodegradable organic nitrogen, mg(TKN-N)/l. Its value is given by Eq. 2.12.

All these effluent fractions are soluble; consequently the total soluble TKN in the effluent (N_t) is given by the sum of these, i.e.

$$N_t = N_a + N_o + N_u \quad (\text{filtered } N_t) \quad (5.28a)$$

The experimental value of N_t will be given by the TKN of the *filtered* effluent sample. If the sample is not filtered the theoretical TKN of the effluent will be greater by the concentration of TKN in the volatile solids, i.e.

$$N_{tv} = N_a + N_o + N_u + f_n \cdot X_{vf} \quad (\text{unfiltered } N_t) \quad (5.28b)$$

where

X_{vf} = effluent volatile solid concentration (mgVSS/l)

f_n = fraction of TKN in VSS $\cong 0.1$ mg(TKN-N)/mgVSS.

5.3 Nitrification capacity

The concentration of nitrate generated in the system (N_c) is given by the TKN in the influent (N_{ti}) less the soluble TKN in effluent (N_t) less the concentration of total influent TKN incorporated in the net sludge generated each day (N_s) i.e.

$$N_c = N_{ti} - N_t - N_s^{**} \quad (5.29)$$

The value of N_s is determined from the mass of nitrogen incorporated in the volatile solids per day. It is given by f_n times the mass of volatile solids wasted per day ($M(X_v)/R_s$), divided by the mean daily flow, i.e.

$$N_s = \frac{f_n \cdot M(X_v)}{R_s \cdot Q} \quad (\text{see Chapter 4, Section 9}).$$

where

$M(X_v)$ is given by Eq. 4.13.

Note that $M(X_v)$ does not include the volatile mass of nitrifiers as the masses of these, we have stated previously, are negligible.

In Eq. 5.29, N_c defines the *nitrification capacity* of the process. A formal definition of nitrification capacity (N_c) is the mass of nitrate produced per unit volume of flow, i.e. mg(NO₃-N)/l of flow.

Eq. 5.29 shows that the nitrification capacity (N_c) is the difference between the influent TKN concentration (N_{ti}) and the sum of the effluent TKN concentration (N_t) and the nitrogen required for sludge production (N_s). The effluent TKN concentration (N_t) depends on the efficiency of nitrification. In calculations, with maximum unaerated sludge mass fraction (f_{xm}), (for a selected sludge age), if the factor of safety (S_f) is selected greater than 1.25 to 1.35 at the *lowest expected temperature* (T_{min}), the efficiency of nitrification always will be high (>90%) and N_t generally will be less than 3 to 4 mgN/l. Also, with $S_f \geq 1.25$ at T_{min} , N_t will be vir-

*The same approach also applied in the design equations for carbonaceous material degradation that, provided $R_s > 3$ days usually one can accept the filtered *biodegradable* COD in the influent as zero (see Chapter 4, Section 3.4).

**This equation presumes that no nitrate is present in the influent and this is true, virtually always, for municipal flows in South Africa.

tually independent of both the process configuration and the subdivision of the sludge mass into aerated and unaerated fractions. *Consequently, for design N_t usually need not be calculated explicitly from Eqs. 5.20, 5.26 to 5.28, but can be taken to be approximately 3 to 4 mgN/l provided that there is reasonable assurance that the actual μ_{nm20} value will not be less than the value accepted for design and that there is sufficient aeration capacity so that nitrification is not inhibited by an insufficient oxygen supply.* Accepting the calculated f_{xm} and selected sludge age (R_s) at the lower temperature, then at higher temperatures the nitrification efficiency and the factor of safety (S_f) both will increase so that at summer temperatures (T_{max}), N_t will be lower, approximately 2 mgN/l.

Dividing Eq. 5.29 by the total influent COD concentration (S_{ti}) yields the nitrification capacity per mgCOD applied to the biological process, N_c/S_{ti} ,

$$N_c/S_{ti} = N_{ti}/S_{ti} - N_s/S_{ti} - N_t/S_{ti} \quad (5.30)$$

where

N_c/S_{ti} = nitrification capacity per mgCOD applied to the process (mgN/mgCOD)

N_{ti}/S_{ti} = TKN/COD ratio of the wastewater

N_s/S_{ti} = nitrogen required for sludge production per mgCOD applied (see Eq. 4.23).

In Eq. 5.30 the nitrification capacity/COD ratio (N_c/S_{ti}) of a process can be estimated approximately by evaluating each of the terms in the RHS of the equation as follows:

N_t/S_{ti} : Provided the constraint for efficient nitrification is satisfied at the lowest temperature (T_{min}) the TKN in the effluent (N_t) can be assumed at T_{min} to be about 4 mgN/l i.e. N_t/S_{ti} will range from 0,005 to 0,01 for S_{ti} ranging from 800 to 400 respectively. At T_{max} , $N_t \cong 2$ mgN/l.

N_s/S_{ti} : This ratio gives the TKN concentration in the influent abstracted for sludge production. Its value will depend on whether settled or raw wastewater is being treated, and the sludge age. The ratio, N_s/S_{ti} , for this purpose can be read from Figure 4.5, or calculated from Eq. 4.23.

N_{ti}/S_{ti} : This value is obtained by measurement of the TKN and COD of the influent and can range from 0,07 for unsettled municipal waste flows to 0,12 for settled flows.

A graphical exposition of the relative importance of these three ratios on N_c/S_{ti} is obtained by plotting N_c/S_{ti} versus sludge age for selected influent TKN/COD (N_{ti}/S_{ti}) ratios ranging from 0,07 to 0,09 for raw wastewater and 0,09 to 0,11 for settled wastewater. The N_c/S_{ti} values were found from Figure 4.5 (or Eq. 4.23) and the N_t/S_{ti} values selected constant at 0,007 (this is for S_{ti} of approximately 600 mgCOD/l). The plots for 14°C and 20°C are shown in Figures 5.8a and 5.8b respectively. Also shown plotted on the diagrams are the minimum sludge ages for unaerated mass fractions of 0,0, 0,2, 0,4 and 0,5, at a selected μ_{nm20} value of

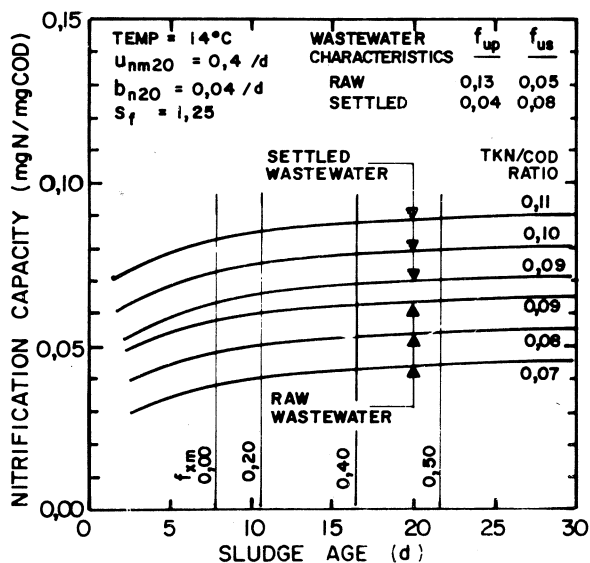


FIGURE 5.8a

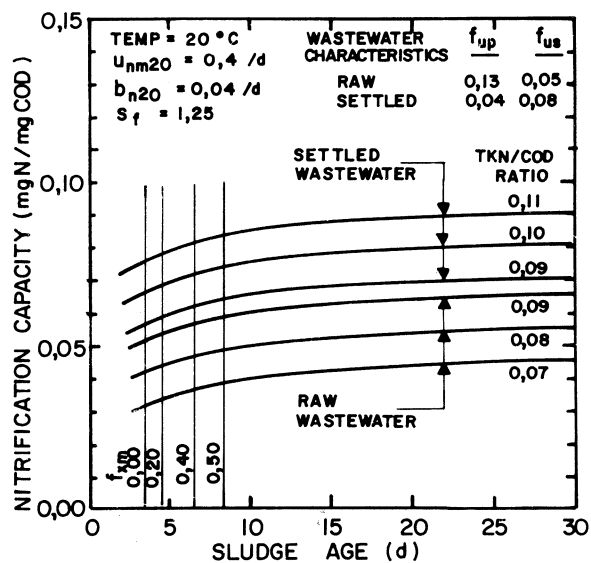


FIGURE 5.8b

FIGURE 5.8: Nitrification capacity per mg COD applied versus sludge age for different influent TKN/COD ratios at 14°C (Figure 5.8a) and 20°C (Figure 5.8b). Also shown are the minimum sludge ages required to obtain complete nitrification at different unaerated sludge mass fractions for $\mu_{nm20} = 0,40/d$.

0.4. For any particular unaerated mass fraction the plotted values of N_c/S_{ti} are valid only at sludge ages greater than the corresponding minimum sludge age. These plots allow one to assess the effects of the three ratios and temperature versus sludge age.

Considering the effects of temperature, to obtain complete nitrification at 14°C (for a selected f_{xm}), the sludge age required is more than double that at 20°C. The corresponding nitrification capacities per influent COD at 14°C show a marginal reduction to those at 20°C (because the sludge production at 14°C is slightly higher than at 20°C).

Considering the effect of sludge age for a selected N_{ti}/S_{ti} value, the N_c/S_{ti} increases as the sludge age increases because the nitrogen needed for sludge production decreases with sludge age and thus becomes available for nitrification. However, the increase is marginal for $R_s > 10$ days.

Considering the influent TKN/COD ratio (N_{ti}/S_{ti}) evidently for both settled and unsettled wastewater, at any selected sludge age, N_c/S_{ti} is very sensitive to the N_{ti}/S_{ti} value; an increase of 0.01 in N_{ti}/S_{ti} causes an increase of about 0.01 in N_c/S_{ti} (0.01 mgN/mgCOD). For the same N_{ti}/S_{ti} ratio for raw or settled wastewater (compare TKN/COD ratio of 0.09 in Figures 5.8a and b), N_c/S_{ti} for unsettled is less than for settled wastewater because more volatile solids are produced per unit COD load from unsettled than from settled wastewater; however, leaving aside this point of difference, the increase in nitrate produced per influent COD will, when the influent TKN/COD ratio increases, increase the difficulties, or, make it impossible, to obtain complete denitrification. This will become clear when denitrification is considered in Chapter 6. As primary settling increases the influent TKN/COD ratio difficulties in obtaining adequate denitrification are always greater with settled than unsettled wastewater; however, because

primary settling removes COD, then for the same flow, the mass of volatile solids and the oxygen demand in the biological process will be lower for settled than for raw wastewater thereby giving rise to savings in plant volume and oxygenation costs.

6. DESIGN EXAMPLE

Design of a nitrification process without denitrification will be considered. When denitrification is required constraints additional to those imposed by nitrification must be applied and these cannot be incorporated in the design until the theory of denitrification is dealt with.

For the purpose of constructive comparison the nitrifying activated sludge plant will be designed for the same waste flows and characteristics as that accepted for the design of a process for carbonaceous material removal (see Chapter 4, Section 12). The wastewater characteristics, for raw and settled wastewaters are listed in Table 4.3 and the additional characteristics needed for nitrification are listed in Table 5.2.

6.1 Effect of nitrification on mixed liquor pH

An important initial consideration is the possible effect of mixed liquor pH on the μ_{nm20} value. In Section 4.3 above it was stated that nitrification consumes alkalinity – 7.14 mg/l as $CaCO_3$ for every 1 mgN/l ammonia converted to nitrate – and if there is insufficient alkalinity in the influent, the mixed liquor pH is depressed below 7.0 causing a reduction in the μ_{nm20} value (see Eq 5.17).

The influent TKN/COD ratio of the raw wastewater is 0.08 mgN/mgCOD (see Tables 4.3 and 5.2). With a relatively low μ_{nm20} value of 0.36/d, the sludge age needs to be 10 days or longer to ensure nitrification

TABLE 5.2 RAW AND SETTLED WASTEWATER CHARACTERISTICS FOR NITRIFICATION (SEE ALSO TABLES 4.3 AND 6.1)

Parameter	Symbol	Value		Units
		Raw	Settled	
Influent TKN concentration	N_{ti}	48	41	mgN/l
Influent TKN/COD ratio	—	0.080	0.114	mgN/mgCOD
Ammonia fraction of TKN	f_{na}	0.75*	0.83*	mgN/mgN
Unbiodegradable soluble organic N fraction	f_{nu}	0.03*	0.04*	mgN/mgN
Nitrogen fraction of volatile solids in influent	f_n	0.10	0.10	mgN/mgVSS
pH of wastewater	—	7.5	7.5	—
Total alkalinity	alk	200	200	mg/l as $CaCO_3$
Maximum specific growth rate of nitrifiers	μ_{nm20}	0.36	0.36	/d.
Influent flow	Q	13.3	13.3	Ml/d

*See Chapter 3, Section 8. Note that settled wastewater characteristics must be selected so that they are consistent with primary sedimentation tank behaviour and the raw wastewater characteristics.

($S_f = 1,3$) at a minimum temperature of 14°C in a purely aerobic process ($f_{xm} = 0,0$) (see Eq. 5.24). At this sludge age the nitrification capacity is about $0,05 \text{ mgN/mgCOD}$ for a TKN/COD ratio of $0,08 \text{ mgN/mgCOD}$, (see Figure 5.8a). Hence the nitrate concentration produced (per ℓ influent) should be about $= 0,05 \cdot 600 = 30 \text{ mgN}/\ell$. This will cause an alkalinity reduction of $7,14 \cdot 30 = 214 \text{ mg}/\ell$ as CaCO_3 . Because the influent alkalinity is only $200 \text{ mg}/\ell$ as CaCO_3 , the mixed liquor alkalinity will drop below $40 \text{ mg}/\ell$ as CaCO_3 causing the mixed liquor pH value to drop below 7 (see Figure 5.6). The low mixed liquor pH value will cause unstable and incomplete nitrification (see Section 4.3 above) and produce an aggressive water (which over a few years can cause considerable damage to the concrete surfaces of the treatment works) and can be the cause of bulking sludges and high solids content in the settling tank effluent due to denitrification action in the secondary settling tank.

The simple approximate calculation above, therefore, can make the designer aware, at an early stage of possible adverse consequences to be expected from a proposed design. In the instance above the designer should give consideration to a nitrification-denitrification process, to recover some of the alkalinity loss due to nitrification, thereby to maintain the pH, reduce the possibility of bulking and eliminate (usually) rising sludge problems in the settling tank. (See Chapter 6, Section 1.4). In general a high TKN/COD ratio coupled with low alkalinity in the influent are reliable indicators warning of problems to be expected in aerobic nitrifying processes.

6.2 Minimum sludge age for nitrification

For the purposes of demonstrating nitrification under purely aerobic conditions, it will be accepted that the influent alkalinity is sufficiently high to maintain an effluent Alkalinity above $40 \text{ mg}/\ell$ as CaCO_3 . Hence no adjustment to μ_{nm20} and K_{n20} for pH needs to be made as the pH is assumed to remain above 7,2. The adjustment of the kinetic constants of nitrification for temperature is given in Table 5.3.

For a completely aerobic process i.e. $f_{xm} = 0$, with $\mu_{nm20} = 0,36/\text{d}$ and $S_f = 1,3$, the minimum sludge age for nitrification (R_{sm}) is found from Eq. 5.24 i.e.

$$\begin{aligned} R_{sm} &= S_f / (\mu_{nmT} - b_{nT}) \\ &= 3,2 \text{ days at } 22^\circ\text{C} \text{ (2,5 days with } S_f = 0,0) \\ &= 8,9 \text{ days at } 14^\circ\text{C} \text{ (7,5 days with } S_f = 0,0) \end{aligned}$$

Clearly, to ensure nitrification throughout the year for the relatively low μ_{nm20} value of $0,36/\text{d}$, the sludge age of a purely aerobic process should be about 10 to 12 days.

6.3 Raw wastewater

The influent TKN concentration of the raw wastewater is $48 \text{ mgN}/\ell$ (see Table 5.2 for wastewater characteristics). Accepting an ammonia fraction of the influent TKN (f_{na}) of $0,75^*$ and an unbiodegradable soluble organic nitrogen fraction (f_{nu}) of $0,03^*$ for the raw sewage (from Table 5.2), gives the influent ammonia concentration (N_{ai}) (from Eq. 2.11) as

$$N_{ai} = 0,75 \cdot 48 = 36 \text{ mgN}/\ell$$

and the unbiodegradable soluble organic nitrogen concentration (N_{ui}), (from Eq. 2.12) as

$$N_{ui} = 0,03 \cdot 48 = 1,50 \text{ mgN}/\ell.$$

Accepting the nitrogen content of the volatile solids in the influent (f_n) as $0,10 \text{ mgN/mgVSS}$ (see Table 5.2), then (from Eq. 2.13) the nitrogen concentration associated with the unbiodegradable particulate COD (N_{pi}) is

$$N_{pi} = 0,10 (0,13 \cdot 600) / 1,48 = 5,3 \text{ mgN}/\ell.$$

Hence (from Eq. 2.14), the biodegradable organic nitrogen concentration in the influent which can be converted to ammonia (N_{oi}) is

$$N_{oi} = 48 (1 - 0,75 - 0,03) - 5,3 = 5,3 \text{ mgN}/\ell.$$

TABLE 5.3 ADJUSTMENT OF KINETIC CONSTANTS OF NITRIFICATION FOR TEMPERATURE

Constant	Symbol	20°C	θ	14°C	22°C	Eq. No.
Max. specific growth rate	μ_{nm20}	0,36	1,123	0,180	0,454	5.15a
Saturation coefficient	K_{n20}	1,0	1,123	0,50	1,26	5.15b
Endogenous respiration rate	b_{n20}	0,04	1,029	0,034	0,042	5.16
Organic N conversion rate	K_r	0,015	1,029	0,013	0,016	5.31

*The actual choice of these values makes relatively little difference to the final results of the nitrification calculations: in the case of the former (f_{na}), most of the organic nitrogen is converted to ammonia in the process and in the case of the latter (f_{nu}), this value merely increases the effluent TKN concentration by a fixed value.

6.4 Settled wastewater

Following the above procedure for settled wastewater, i.e. $f_{na}=0,83$, $f_{nu}=0,04$ (see Table 5.2) yields the following:

$$N_{ti} = 41,0 = 41,0 \text{ mgN/l}$$

$$N_{ai} = 0,83 \cdot 41,0 = 34,0 \text{ mgN/l}$$

$$N_{ui} = 0,04 \cdot 41,6 = 1,50 \text{ mgN/l}$$

$$N_{pi} = 0,10 (0,04 \cdot 360) / 1,48 = 1,0 \text{ mgN/l}$$

$$N_{oi} = 41,0 - 34,0 - 1,5 - 1,0 = 4,5 \text{ mgN/l}$$

6.5 Nitrification process behaviour

For any sludge age longer than about 2 days the effluent biodegradable organic nitrogen concentration (N_{oe}) is given (from Eq. 5.26)

$$N_{oe}^* = N_{oi} / (1 + K_{RT} X_a R_{hn}) \quad (\text{mgN/l})$$

where from Eq. 4.9a, $R_{hn} X_a = M(X_a) / Q$. Hence

$$N_{oe} = N_{oi} / (1 + K_{RT} M(X_a) / Q) \quad (\text{mgN/l}) \quad (5.31)$$

From Eq 5.27, the unbiodegradable organic nitrogen in the effluent is

$$N_{ue} = N_{ui} = 1,50 \text{ mgN/l for raw and settled wastewater} \quad (5.32)$$

The nitrogen required for sludge production (N_s) is given by Eq 4.23 i.e.

$$N_s = f_n \left\{ \frac{(1 - f_{us} - f_{up}) Y_h}{(1 + b_{HT} R_s)} + \frac{f_{up}}{f_{cv}} \right\} S_{ti} \quad (5.33)$$

The influent TKN concentration available for nitrification (N'_{ti}) is the difference between the total TKN concentration in the influent (N_{ti}) and that required for sludge production (N_s) (from Eq. 5.19)

$$N'_{ti} = N_{ti} - N_s \quad (5.34)$$

If the sludge age of the process is longer than the minimum required for nitrification ($R_s > R_{sm}$ for $S_f = 1,00$, Eq. 5.14), the TKN concentration available for nitrification will be partially or almost wholly nitrified to nitrate and the effluent nitrate concentration (N_{ne}) is the difference between the nitrogen available for nitrification (N'_{ti} , Eq. 5.34) and the effluent TKN concentration (N_{te}). If $R_s < R_{sm}$, no nitrification takes place; the effluent nitrate concentration (N_{ne}) is zero and the effluent ammonia concentration (N_{ae}) is equal to the nitrogen available for nitrification (N'_{ti} , Eq. 5.34) minus the sum of the unbiodegradable and biodegradable TKN concentrations in the effluent ($N_{ue} + N_{oe}$). For both $R_s < R_{sm}$ and $R_s > R_{sm}$, the effluent TKN concentration (N_{te}) is the sum of effluent unbiodegradable and biodegradable organic nitrogen concentrations (N_{ue} and N_{oe} respectively) and the effluent ammonia concentration (N_{ae}).

For $R_s > R_{sm}$:

From Eq 5.13, the effluent ammonia concentration is given by

$$N_{ae} = K_{nT} (b_{nT} + 1/R_s) / \{\mu_{nmT} - (b_{nT} + 1/R_s)\} \quad (5.35a)$$

Hence the effluent TKN concentration (N_{te}) is

$$N_{te} = N_{ae} + N_{oe} + N_{ue} \quad (\text{mgN/l}) \quad (5.36a)$$

and the effluent nitrate concentration (N_{ne}) is

$$\begin{aligned} N_{ne} &= N'_{ti} - N_{te} \quad (\text{mgN/l}) \\ &= N_{ti} - N_s - N_{te} \quad (\text{from Eq. 5.34}) \end{aligned} \quad (5.37a)$$

Analogous to the concentration of active heterotrophic organisms (see Eq. 4.10), the nitrifier organism mass is given by

$$M(X_n) = M(N_{ne}) Y_n R_s / (1 + b_{nT} R_s) \quad (\text{mgVSS}) \quad (5.38a)$$

where

$$\begin{aligned} M(N_{ne}) &= \text{mass of nitrate generated per day} \\ &= Q \cdot N_{ne} \end{aligned}$$

where N_{ne} is given by Eq. (5.37a).

The oxygen demand for nitrification is simply 4,57 mgO/mgN times the mass of nitrate produced per day, i.e.

$$M(O_n) = 4,57 M(N_{ne}) \quad (\text{mgO/d}) \quad (5.39a)$$

For $R_s < R_{sm}$:

For $R_s < R_{sm}$, no nitrification takes place. Hence the effluent nitrate concentration is zero i.e.

$$N_{ne} = 0,00 \quad (\text{mgN/l}) \quad (5.37b)$$

The effluent ammonia concentration is

$$\begin{aligned} N_{ae} &= N'_{ti} - N_{oe} - N_{ue} \quad (\text{mgN/l}) \\ &= N_{ti} - N_s - N_{oe} - N_{ue} \quad (\text{from Eq. 5.34}) \end{aligned} \quad (5.35b)$$

The effluent TKN concentration is

$$N_{te} = N_{ae} + N_{ue} + N_{oe} \quad (5.36b)$$

The nitrifier sludge mass and the nitrification oxygen demand are zero, i.e.

$$M(X_n) = 0,0 \quad (\text{mgVSS}) \quad (5.38b)$$

$$M(O_n) = 0,0 \quad (\text{mgO/d}) \quad (5.39b)$$

Substituting the respective influent nitrogen concentrations for raw and settled wastewaters and the values of the kinetic constants at 14°C into Eqs. 5.31 to 5.39 and plotting the results at different sludge ages yields Figure 5.9. In Figure 5.9a, the different concentrations of nitrogen in the process versus sludge age for raw and settled wastewater at 14°C are shown. In Figure 5.9b the nitrifier sludge mass and nitrification oxygen demand for raw and settled wastewater at 14°C are shown. Also shown in Figure 5.9b are the carbonaceous and total oxygen demands for raw and settled wastewater at 14°C. The calculations were repeated for 22°C and are shown in Figure 5.10a and b.

*The subscript e used here and later refers to the effluent. However, $N_o = N_{oe}$, $N_a = N_{ae}$, etc. because completely mixed conditions are assumed.

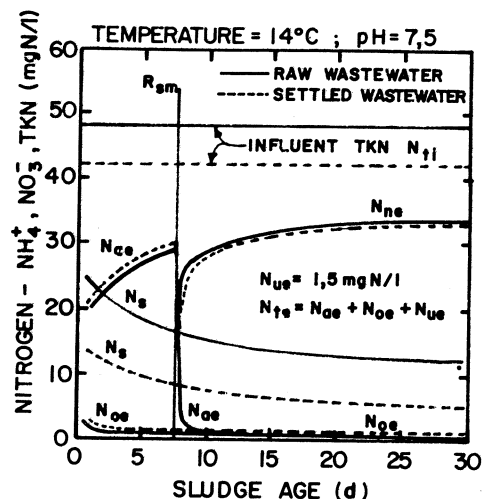


FIGURE 5.9a

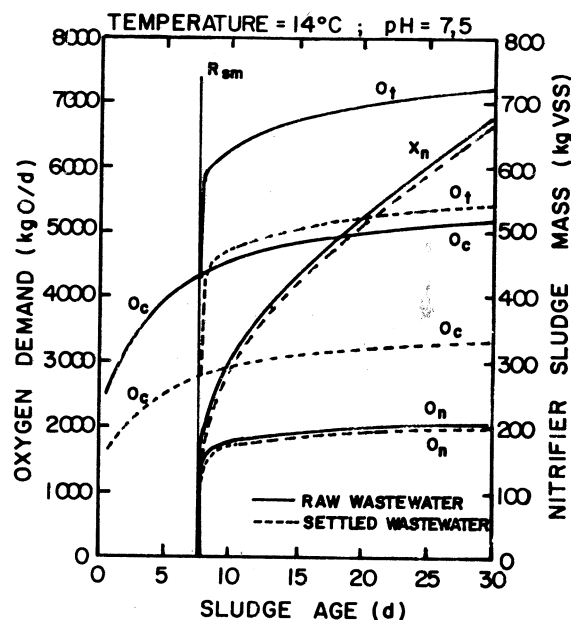


FIGURE 5.9b

FIGURE 5.9: Effluent ammonia (N_{ae}), organic nitrogen (N_{oe}), and nitrate (N_{ne}) concentrations and nitrogen required for sludge production (N_s) (Figure 5.9a), carbonaceous (O_c), nitrification (O_n), and total (O_t) oxygen demand and the nitrifier sludge mass (X_n) (Figure 5.9b) versus sludge age for raw and settled wastewater at 14°C.

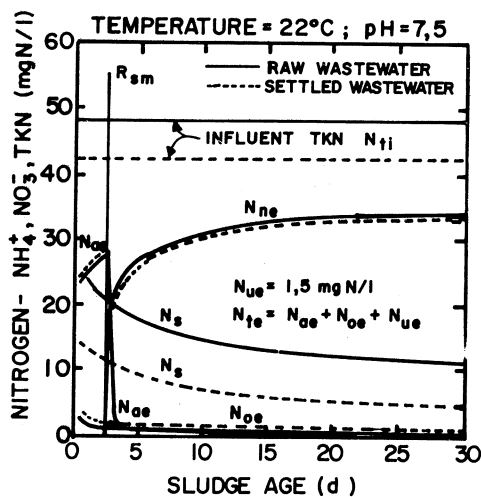


FIGURE 5.10a

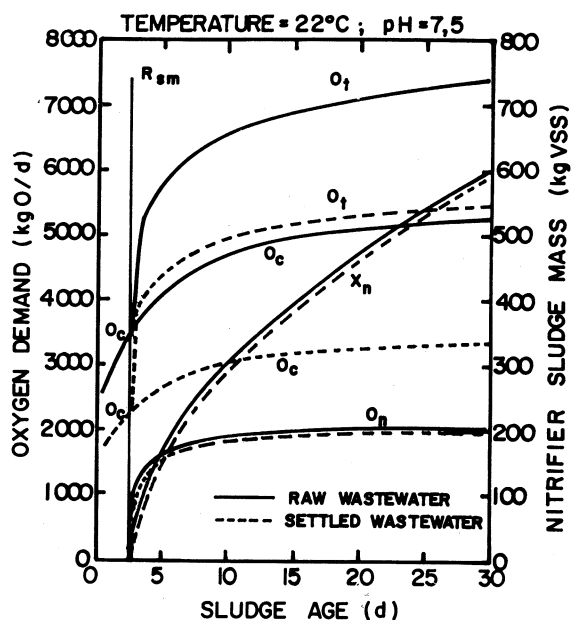


FIGURE 5.10b

FIGURE 5.10: Effluent ammonia (N_{ae}), organic nitrogen (N_{oe}), and nitrate (N_{ne}) concentrations, nitrogen required for sludge production (N_s) (figure 5.10a) carbonaceous (O_c), nitrification (O_n), and total (O_t) oxygen demand and the nitrifier sludge mass (X_n) (Figure 5.10b) versus sludge age for raw and settled wastewater at 22°C.

Figures 5.9 and 5.10 show that once the sludge age is approximately 25 per cent longer than the minimum required for nitrification, nitrification is complete and comparing the results for raw and settled wastewater, there is little difference between the nitrification oxygen demand and the concentrations of ammonia, nitrate and TKN in the effluent. The reasons for this similar behaviour are: (1) the primary settling tank removes only a small fraction of the influent nitrogen*; and (2) settled wastewater results in lower sludge production, so that the nitrogen available for nitrification in raw and settled wastewater are approximately equal. Temperature has little effect on the different concentrations of nitrogen. However, a change in temperature causes a significant change in the minimum sludge age for nitrification.

Considering Figures 5.9a and 5.10a, the effluent organic nitrogen concentration (N_{oe}) is less than 2 mgN/l for all sludge ages greater than about 3 days. For sludge ages about 25 per cent longer than the minimum for nitrification, the effluent ammonia concentration (N_{ae}) is less than 1 mgN/l so that for $R_s > 1.25 R_{sm}$ the effluent TKN concentration can be taken to be approximately 4 mgN/l. The increase in nitrate concentration (N_{ne}) with increase in sludge age is principally attributable to the reduction in the nitrogen required for sludge production (N_s). This is important for nutrient removal plants as it indicates that increasing the sludge age of a process, increases the nitrification capacity (see section 5.2 above) even though the effluent TKN concentration remains constant at about 4 mgN/l (see Figure 5.8). For $R_s < R_{sm}$, the effluent ammonia concentration (N_{ae}) and hence the effluent TKN concentration (N_{te}), increases with increasing sludge age up to R_{sm} because N_s reduces for increases in R_s .

Figures 5.9b and 5.10b show that the nitrification oxygen demand increases rapidly once $R_s > R_{sm}$ but if $R_s > 1.25 R_{sm}$, the increase is marginal irrespective of the temperature or wastewater type, i.e. between sludge ages 10 and 30 days about 2000 kgO/d are required for nitrification. This nitrification oxygen demand represents an increase of 40% and 60% respectively** above the carbonaceous oxygen demand for the raw and settled wastewater. However, the total oxygen demand for treating settled wastewater is only 75% of that for treating raw wastewater.

In order that nitrification can proceed without inhibition by oxygen limitation, it is important that the aeration equipment is adequately designed to supply the total oxygen demand; generally heterotrophic organism growth takes precedence over nitrifier growth when oxygen supply becomes insufficient. This is because heterotrophic organisms can grow adequately with dissolved oxygen concentrations of 0.5 to 1.0 mgO/l whereas nitrifiers require a minimum concentration of 1 to 2 mgO/l.

The nitrifiers sludge mass given in Figures 5.9b and 5.10b shows a rapid increase once $R_s > R_{sm}$ and is ap-

proximately the same for raw or settled sewage, but is slightly higher at 14°C than at 22°C. Comparing the nitrifier sludge mass to the heterotrophic sludge mass, even at high TKN/COD ratios, the nitrifier sludge mass comprises only about 2 per cent of total volatile mass and may be neglected in the determination of the volatile solids concentration in the activated sludge process treating domestic wastewaters.

It is worth repeating that primary sedimentation removes only a minor fraction of the TKN but a significant fraction of COD. Even though the settled wastewater has a lower TKN than the raw wastewater the process effluent nitrate concentration will not reflect this difference because the nitrogen removed for sludge production is lower for settled than for raw wastewater. Consequently the nitrate generated is approximately the same for both waste flows. In contrast, the maximum denitrification possible (denitrification potential) in the process is dependent principally on the influent COD concentration and this concentration, we have seen, is significantly reduced by primary sedimentation. This may result in a situation where it may be possible to obtain complete nitrate removal when treating raw wastewater but highly unlikely when treating settled wastewater. This difference in removal can have a profound effect on the process design for biological excess phosphorus removal.

9. REFERENCES

- ARKLEY, M. and MARAIS, G.v.R. (1982) Effect of the anoxic sludge mass fraction on the activated sludge process. Research Report No W 38, Dept. of Civil Eng., Univ. of Cape Town.
- DOLD, P.L., EKAMA, G.A. and MARAIS, G.v.R. (1980) A general model for the activated sludge process. *Prog. Wat. Tech.*, **12**, 47-77.
- DOWNING, A.L., PAINTER, H.A. and KNOWLES, G. (1964) Nitrification in the Activated Sludge Process, *J. Proc. Inst. Sew/Purif.*, **64**, 2, 130-158.
- EKAMA, G.A., VAN HAANDEL, A.C. and MARAIS G.v.R. (1979) The present status of research on nitrogen removal: a model for the modified activated sludge process. Presented at the Symposium on nutrient removal, SA Branch of the IWPC, Pretoria, May 1979. Proc. published by Water Research Commission of South Africa.
- HERBERT, D. (1958) Recent progress in microbiology. VII. International congress on microbiology. Ed. by Tunevall, Alinquist and Wiksell, Stockholm, 1958, 381-396.
- LOEWENTHAL, R.E. and MARAIS, G.v.R. (1977) Carbonate chemistry of aquatic systems: Theory and application. Ann Arbor Science, Michigan, USA.
- LOVELESS, J.E. and PAINTER, H.A. (1968) The influence of metal ion concentration and pH value in the growth of a *Nitrosomonas* strain isolated from activated sludge. *J. Gen. Micro.*, **52**, 1-14.
- MALAN, W.N. and GOUWS, E.P. (1966) Geaktiveerde slyk vir rioolwatersuiwering op Bellville. Research Report, Council for Scientific and Industrial Research, Nov. 1966.
- MONOD, J. (1950) Technique of continuous culture - Theory and application, (Translation from French), *Ann. Inst. Pasteur*, **79**, 167.
- NEYTZEL-DE WILDE, F.G. (1977) Treatment of effluents from ammonia plants - Part 1. Biological nitrification of an inorganic effluent from a nitrogen-chemicals complex. *Water S.A.*, **3**(3) 113-122.

*This depends on the state of the sewage; if the sewage is "fresh" a larger fraction of the TKN can still be in particulate proteinaceous form.

**Roughly for $R_s > 1.25 R_{sm}$, the increase in the total oxygen demand due to nitrification is given by $M(O_t) = (1 + 5 f_{ns})M(O_c)$ where f_{ns} is the TKN/COD ratio of the wastewater. For processes including denitrification see Chapter 6, Section 7.6.2.

STENSTROM, M.K. and PODUSKA, R.A. (1980) The effect of dissolved oxygen concentration on nitrification. *Wat. Research*, 14, 6, 643-649.

VAN HAANDEL, A.C. and MARAIS, G.v.R. (1981) Nitrification and denitrification in the activated sludge process. Research Report W 42, Dept. of Civil Eng., Univ. of Cape Town.
